# AI AIDED OPPORTUNITY SUB-SPACE IDENTIFICATION FOR PHARMA ASSETS

**Dissertation Title:**

**AI AIDED OPPORTUNITY SUB-SPACE IDENTIFICATION FOR PHARMA ASSETS**

**(Molecule-level opportunity assessment at scale using open data)**

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## Abstract

The pharmaceutical industries are moving from “branded house” to “house of brands” model where molecules must support commercial cases. The essential part of this is to assess the commercial potential for many assets with consistent and rapid output.

This judgement is typically made through an opportunity assessment: a grid that divides an indication’s patient population into mutually exclusive and collectively exhaustive segments. It is built in four steps:

- **Segment:** define the segment dimensions.
- **Size:** quantify each segment.
- **Align:** prioritize the segments.
- **Associate:** link the priority segments to the care settings and prescribers that serve them.

As per the request, strategist and analyst perform this entire process manually and also one indication at a time, which takes weeks for a each asset. This results in few indications per molecule mapped and the opportunity funnels seems narrow.
The dissertation proposes an AI System that automated the opportunity assessment for the indication for a particular molecule, at the portfolio level, using open available internet data

For each molecule, the AI works through five Steps:

1. **Discover:** Identify candidate indication from mechanism, same-mechanism analogues and literature, using ChEMBL, Open Targets and PubMed.
2. **Segment:** Split indication into segment dimensions such as age band, risk group, line of therapy or care setting, and geography, using LLM extraction from ClinicalTrials.gov and public product labels that includes (EMA, FDA and Openlabel data)
3. **Size:** Sizing the segment with an epidemiology funnel (prevalent, diagnosed, treated) using IHME disease burden data, WHO and CDC data.
4. **Align:** Sort segments into Primary, Secondary, Spillover and Not-a-Target tiers. A viability score is used on unmet need, addressable population, competitive intensity, scientific rationale, development feasibility and momentum. These are ranked by LightGBM Learning-to-rank model explained with SHAP.
5. **Associate:** Link the priority segments to prescriber specialties and care setting by public data such as CMS public prescriber and NPI registry data.

Each segment is supported by an evidence brief with source citations, produced through hybrid retrieval and Graph-RAG over a biomedical knowledge graph. Evaluation: A time-split back-test uses evidence up to 2019 to predict which molecule–indication–segment opportunities entered Phase 2/3 trials or were approved in 2020–2024. Further checks cover:

- **Extraction accuracy:** entity-level F1.
- **Evidence faithfulness:** RAGAS.
- **Sizing accuracy:** versus published estimates.
- **Segmentation validity.**
- **Blinded expert review.**
- **Throughput:** versus manual assessment.

The outcome is a reproducible, open-data prototype that widens the opportunity funnel with ranked, sized, cited and explainable draft assessments for analysts to validate.

**Keywords:** *Opportunity Assessment, Patient Segmentation, Opportunity Identification, Large Language Models, Knowledge Graphs, Retrieval-Augmented Generation, Learning-to-Rank, Explainable AI, Open Data*

## 1. Broad Area of Work

Applied Artificial Intelligence and pharmaceutical commercial strategy is the base for this dissertation. This compressed of LLMs, knowledge graphs, RAG and machine-learning-based prioritization to automate the opportunity-assessment process for many molecules at once, using open source data. The work draws on the following AI areas:

- **Biomedical Natural Language Processing:** extracting structured entities (molecule, target, disease, population, age band, risk factor, line of therapy, care setting, geography, comparator, endpoint) from public trial registries, abstracts and product labels [5], [6].
- **Knowledge representation and Graph-RAG:** linking molecules, targets, diseases, segments, trials, publications, burden estimates and prescribers in a knowledge graph, and grounding LLM output in cited public sources [2], [3].
- **LLM-assisted segmentation:** proposing segment dimensions and categories that satisfy segmentation rules, checked by an LLM critic and rule-based validators [1], [4].
- **Machine learning for prioritization:** scoring and ranking segments with multi-criteria and learning-to-rank models to assign priority tiers [16].
- **Explainable and trustworthy AI:** evidence citations, SHAP attributions and faithfulness checks so that every segment cell can be audited [15], [18].

# 2. Background

## 2.1 Business context

A pharmaceutical asset is usually described by its approved or targeted indication, but its value is concentrated in narrower opportunity sub-spaces: combinations of indication, patient segment, line of therapy or care setting, and geography where unmet need is meaningful, the population is sizeable and competition leaves room to win. The move from a branded-house to a house-of-brands operating model means each molecule must establish its own commercial case, repeatedly, as evidence changes. Organizations therefore need to assess many molecules at scale and with pace, and to widen the opportunity funnel before costly manual validation begins.

## 2.2 Opportunity assessment and its four steps

An opportunity assessment is a segmentation-based view of the treated or treatable population for one indication. It arranges patients into a grid across two or more dimensions that are mutually exclusive and collectively exhaustive (MECE) [1], anchors the grid to a sized population, and overlays priorities showing where to focus. It is a segmentation, not a patient-journey diagram, forecast or marketing funnel. It is built in four steps:

| Step | Question answered | Typical inputs |
|---|---|---|
| 1. Segment | How does the population split by clinical and treatment characteristics? | Labels, guidelines, trial eligibility criteria, literature |
| 2. Size | How many patients sit in each segment? | Epidemiology: prevalence → diagnosed → treated funnel |
| 3. Align | Which segments are Primary, Secondary, Spillover or Not-a-Target? | Unmet need, competition, evidence, feasibility |
| 4. Associate | Which care settings and prescribers serve the priority segments? | Prescriber and provider data by specialty |

Each dimension must be meaningful (it changes strategy), identifiable (it can be found in data or a credible proxy) and actionable (it changes targeting, messaging or channel), with no more than three dimensions per axis. Common dimensions include age band, risk group, severity, biomarker status, line of therapy, treatment status, care setting and geography.

## 2.3 Current practice and its limitations

Opportunity assessments are built manually by analysts who read labels, guidelines, trials, literature and epidemiology. The process takes several weeks per indication, is hard to refresh, varies between analysts and tends to favour segments a team already knows. Because it starts only after an indication has been chosen, it cannot help decide which indications of a molecule deserve an assessment. The evidence base also grows faster than it can be read: ClinicalTrials.gov holds more than 500,000 registered studies [13] and PubMed indexes over 36 million citations. General-purpose LLMs can suggest segmentations from a prompt [7], but

without retrieval they produce uncited and sometimes incorrect structures, do not size segments from data and do not rank opportunities.

### 2.4 Relevance to the employing organization

The student works in commercial and go-to-market analytics, supporting forecasting, launch readiness, market-entry strategy and scientific-leader analytics. Opportunity assessments are a direct upstream input to this work. A method that produces sized, prioritised and evidence-backed draft assessments across many molecules would shorten the assessment cycle and make it repeatable. The dissertation itself uses only open public data and is independent of any internal data, tools or pipeline information.

### 2.5 Research gap and research questions

AI research in pharma focuses mainly on molecular problems such as target–disease link prediction for drug repurposing [8], [9], or on general LLM question answering [7]. Repurposing models estimate biological plausibility but ignore segmentation and commercial factors, and LLMs summarize well but hallucinate [2], [7]. There is little published work that automates the full opportunity-assessment process – segment, size, align and associate – from open data, grounds each segment in citations, works across a portfolio and is validated against later outcomes.

- **Q1:** Can a retrieval-grounded LLM pipeline generate valid (MECE, meaningful, identifiable, actionable) opportunity segmentations for molecule-indication pairs from open data?
- **Q2:** How accurately can segments be sized from open epidemiology sources compared with published estimates?
- **Q3:** Does a learned viability score prioritize segments that later advanced better than heuristic, single-signal and LLM-only baselines?
- **Q4:** How much faster is the system than manual assessment, and what share of later-validated opportunities appears in a shortlist small enough for manual review?

## 3. Objectives

1. To define an opportunity sub-space schema (Molecule × Indication × Patient segment × Line of therapy / Care setting × Geography) aligned to the four opportunity-assessment steps, and to formalize segmentation rules as machine-checkable constraints.
2. To build an indication-discovery engine that expands each molecule into plausible indications through mechanism, analogue and literature routes using open pharmacology data.
3. To build an LLM extraction pipeline that converts public trial records, abstracts and labels into structured entities, benchmarked against a fine-tuned biomedical transformer.
4. To generate patient segmentations (Segment) for each indication and validate them against segmentation rules and expert judgement.
5. To size each segment (Size) from an open epidemiology funnel and to report sizing confidence.
6. To prioritize segments (Align) with a multi-criteria baseline and a learning-to-rank model explained with SHAP, assigning Primary, Secondary, Spillover and Not-a-Target tiers.
7. To link priority segments to prescriber specialties and care settings (Associate) using open public provider data.
8. To evaluate the system through time-split back-testing, sizing accuracy, ablation, blinded expert review and a scale study, and to deliver a working prototype.

## 4. Scope of Work

The scope follows the guideline of one problem, one data domain, one core approach and a focused set of experiments. All design, data engineering, modelling, evaluation and reporting will be carried out by the student independently.

### 4.1 In scope

- **Unit of analysis:** individual molecules (marketed and clinical-stage), characterized by mechanism, target and modality.
- **Portfolio sample:** 50–100 publicly documented molecules across three to four therapeutic areas (for example vaccines and infectious disease, oncology, cardiometabolic and immunology).
- **Geography:** United States as the primary market (richest open data), with selected ex-US markets sized from global sources and flagged with a confidence level.
- **Data:** entirely open and publicly available internet sources (Section 5.3). No internal, confidential, licensed or proprietary data, tools, templates or pipeline information will be used.
- **Models:** open-weight LLMs (for example Llama or Mistral families) or publicly available LLM APIs, open-source biomedical transformers (PubMedBERT / Bio BERT) and open-source libraries.
- **Prototype:** a Python application with a Streamlit interface offering a portfolio view and an opportunity-assessment drill-down per molecule–indication.

### 4.2 Planned experiments and evaluation metrics

| Experiment | Comparison | Metrics |
|---|---|---|
| E1: Extraction | Zero-/few-shot LLM vs fine-tuned PubMedBERT / BioBERT on an annotated gold set | Entity-level precision, recall, F1; Cohen's κ |
| E2: Retrieval & grounding | BM25 vs dense vs hybrid (RRF) vs Graph-RAG | Recall@k, MRR; RAGAS faithfulness, context precision |
| E3: Segmentation validity | LLM-only vs retrieval-grounded segmentation | MECE rule pass rate; expert rating of dimensions; Krippendorff's α |
| E4: Size accuracy | System-derived segment sizes vs published prevalence / treated-population estimates | Median absolute percentage error; share within ±25% |
| E5: Align (back-test) | Random, Open Targets only, LLM-only, heuristic, LambdaMART; evidence ≤ 2019 predicting Phase 2/3 starts or approvals in 2020–2024; ablation by dimension | Recall@k (primary), Precision@k, NDCG@k; change on ablation |
| E6: Scale & pace | System vs manual assessment of the same indications | Indications mapped per hour, cost per molecule, analyst time saved |

### 4.3 Out of scope

- Revenue forecasting, pricing and financial valuation (e.g., NPV).
- Prediction of clinical efficacy or safety, and molecular design.
- Final, commercially validated opportunity assessments; outputs are drafts for analyst validation.
- Any internal or licensed data (e.g., proprietary claims, sales or CRM data), production deployment and enterprise integration.

## 5. Methodology

### 5.1 Research design

The study follows a design-science approach: an artefact (the opportunity-assessment System) is designed, built and evaluated against explicit baselines. The System mirrors the four opportunity-assessment steps, preceded by indication discovery and supported throughout by an evidence layer:

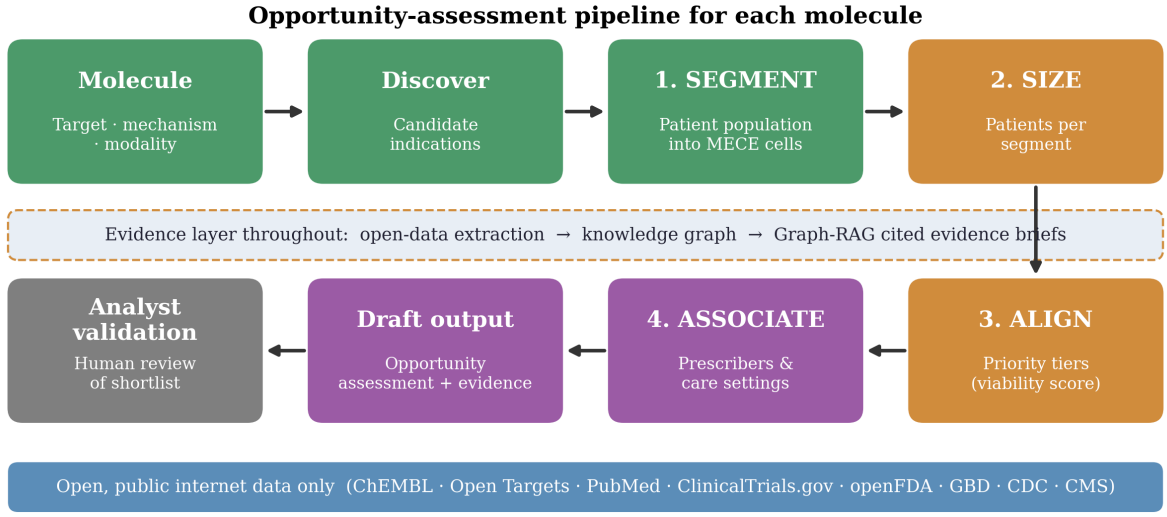



*Figure 1: Research design – the four opportunity-assessment steps (Segment, Size, Align, Associate) applied to each molecule.*

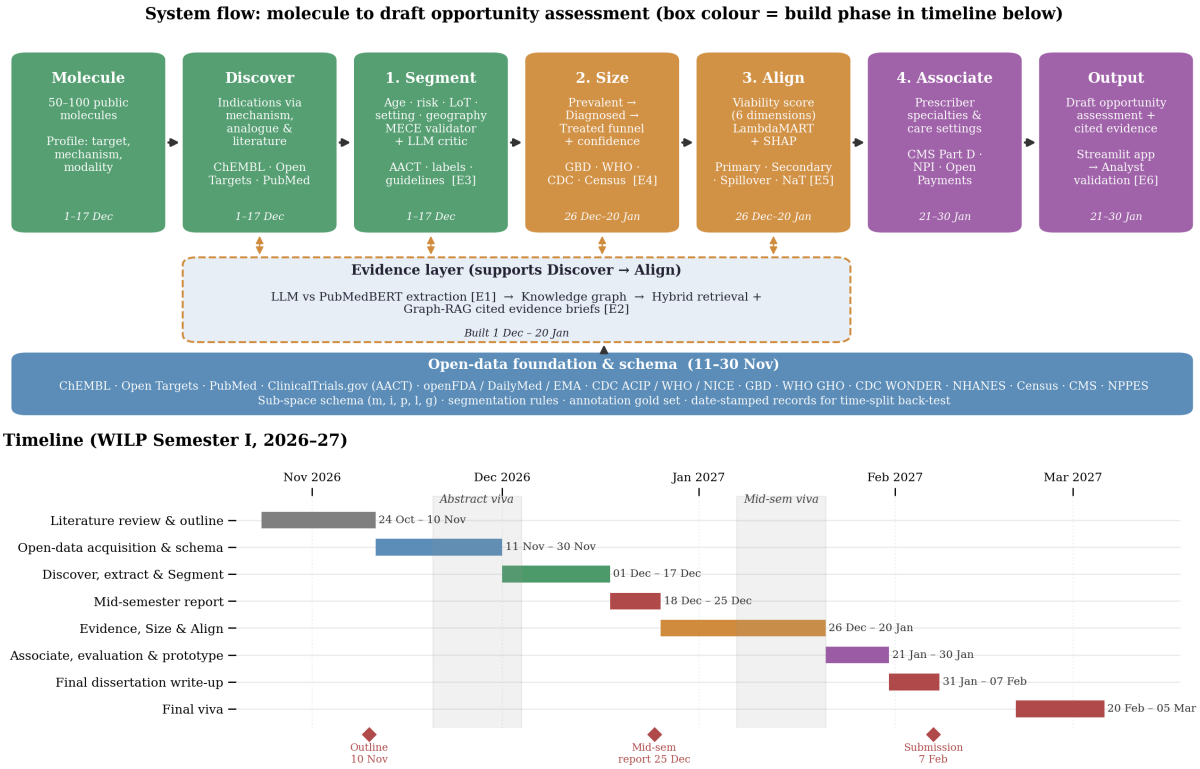
Because the System sits at the top of the funnel, it is optimized first for recall (not missing good opportunities) and then for a shortlist small enough to validate (precision at k).

Figure 2 shows the end-to-end system flow together with the timeline in which each component will be built; box colors in the flowchart correspond to the phases in the timeline.

### 5.2 Problem formulation

An opportunity sub-space is a tuple s = (m, i, p, l, g): molecule m, indication i, patient segment p, line of therapy or care setting l, and geography g. For each molecule–indication pair the System produces an opportunity assessment M(m, i) = {cells s} whose segments satisfy the MECE constraint, a size n(s) for each cell, a viability score f(s) and a priority tier, and a set of

associated prescriber specialties and care settings. Ranking is performed within each molecule (query groups) and normalised across molecules for the portfolio view.




**Timeline (WILP Semester I, 2026-27)**




*Figure 2: System flowchart integrated with the plan of work (Section 6). [E1]–[E6] mark where each experiment is run.*

## 5.3 Open data sources by opportunity-assessment step

| Step | Open source | Use |
|---|---|---|
| Discover | ChEMBL [11], PubChem | Mechanism, target, modality, known indications, maximum phase |
| Discover | Open Targets Platform [10] | Target–disease association scores (scientific rationale) |
| Discover / Segment | PubMed (NCBI E-utilities), Europe PMC | Literature evidence on mechanism, populations and unmet need |
| Segment | ClinicalTrials.gov (AACT database) [13] | Conditions, eligibility (age, risk, line of therapy), sites, phases, sponsors, dates |
| Segment / Align | openFDA, Drugs@FDA, DailyMed, EMA EPARs | Approved indications, populations and dates; competitor coverage |
| Segment | Public clinical guidelines and immunisation schedules (e.g., CDC ACIP, WHO SAGE, NICE) | Recognised severity, risk-group and line-of-therapy definitions |
| Size | Global Burden of Disease (IHME GBD Results Tool) [14] | Incidence, prevalence, DALYs, deaths by age, sex and country |
| Size | WHO Global Health Observatory, CDC WONDER, NHANES, U.S. Census | Population denominators, diagnosis and treatment rates |

| Step | Open source | Use |
|---|---|---|
| Align | AACT and openFDA (as above) | Competitive intensity, development feasibility, momentum |
| Associate | CMS Medicare Part D Prescribers by Provider and Drug; NPPES NPI Registry; CMS Open Payments | Prescriber specialties, volumes and care settings linked to segments |

All sources are freely downloadable or accessible through public APIs. Conditions are normalised to MeSH / ICD-10 / EFO, geographies to ISO-3166, ages to standard bands, molecules to ChEMBL IDs and prescribers to NPI taxonomy codes. Every record keeps its source identifier and date so that time-split evaluation can be enforced.

### 5.4 Evidence layer: extraction, knowledge graph and Graph-RAG

**Extraction (E1).** A gold set of about 200–300 trial records, abstracts and label sections is annotated with written guidelines; a subset is double-annotated for Cohen's κ. Method A uses zero- and few-shot LLM extraction into a fixed JSON schema, generated record by record to avoid output truncation. Method B fine-tunes PubMedBERT [6] / BioBERT [5] for named-entity recognition on 70% of the gold set. Systemh are evaluated on the held-out 30% with exact- and partial-match precision, recall and F1.

**Knowledge graph.** The knowledge graph links the entities and evidence used across the pipeline, as follows:

- **Nodes:** Molecule, Target, Disease, Segment, Setting / LoT, Geography, Trial, Publication, Label, Burden Estimate and Prescriber Specialty.
- **Edges:** include
  - Molecule Act_on [Target]
  - Target Associated with [Disease]
  - Trial Enroll [Segment]
  - Segment Has Burden [BurdenEstimate]
  - Segment Treated by [PrescriberSpecialty]
- **Provenance:** Every edge carries a source ID and date.
- **Design and implementation:** the design follows the heterogeneous biomedical networks used for repurposing [8]; the graph might be built in NetworkX (Neo4j optional).

**Retrieval and evidence briefs (E2).** BM25, dense biomedical embeddings, a reciprocal-rank-fusion hybrid and Graph-RAG [3] are compared within a retrieval-augmented generation setup [2]. For each segment cell the LLM writes an evidence brief in which every claim cites an NCT ID, PMID, label or dataset; uncited claims are removed. Metrics: Recall@k, MRR, RAGAS faithfulness and context precision [15].

### 5.5 Stage 0 – Molecule profiling and indication discovery

Each molecule is profiled by target, mechanism, modality, approved indications and highest phase. Candidate indications are generated through three routes and merged: (a) mechanism – diseases associated with the molecule's targets in Open Targets [10] above a tuned threshold,

using ChEMBL [11] for mechanism and target mapping; (b) analogue – indications pursued by molecules sharing the mechanism, following drug-repurposing practice [8], [9]; (c) literature – diseases co-mentioned with the molecule or target in PubMed.

### 5.6 Stage 1 – Segment: patient segmentation (E3)

For each molecule–indication pair, the LLM proposes up to three candidate segmentations using retrieved evidence: dimensions (e.g., age band × risk group, or line of therapy × biomarker status), their categories and the layout (two-dimensional crosstab, hierarchical or nested). Dimensions are restricted to those observed in trial eligibility criteria, labels or guidelines. A rule-based validator checks that categories are mutually exclusive and collectively exhaustive and that there are at most three dimensions per axis, consistent with segmentation theory [1]; an LLM critic, run as a reason–act loop [4], checks the meaningful–identifiable–actionable test. Failing structures are revised or discarded. Output: a structured JSON grid per indication.

### 5.7 Stage 2 – Size: epidemiology funnel (E4)

Each cell is sized through a population funnel anchored on open data:

$$Population(g, age) \rightarrow Prevalent \rightarrow Diagnosed \rightarrow Treated \rightarrow Segment\ share$$

Population and prevalence come from GBD [14], WHO and Census data; diagnosis and treatment rates and segment shares come from CDC sources, NHANES and published literature retrieved by the evidence layer. Each size carries a confidence level (High / Medium / Low) based on source type and directness. Sizes are compared with independently published estimates to measure error.

### 5.8 Stage 3 – Align: viability scoring and priority tiers (E5)

| Dimension | Example open-data features |
|---|---|
| Scientific rationale | Open Targets score; supporting publications; analogue approvals |
| Unmet need | Burden not covered by approved therapies; number of approved options for the segment |
| Addressable population | Segment size from Stage 2 (log-scaled) |
| Competitive intensity | Approved competitors and active late-phase trials in the segment (inverse) |
| Development feasibility | Highest phase reached; historical phase-transition success for the class [12] |
| Momentum | Year-on-year trial starts, new sponsors, publication growth |

Baseline: a weighted multi-criteria score with min–max normalization and sensitivity analysis on the weights:

$$Score(s) = \Sigma\, w_k \cdot \tilde{x}_k(s), \quad \Sigma\, w_k = 1$$

Learned model: LightGBM LambdaMART [16] with molecules as query groups, tuned by grouped cross-validation. Score thresholds assign cells to Primary, Secondary, Spillover and Not-a-Target tiers. SHAP values [18] explain each cell's tier alongside its evidence brief.

### 5.9 Stage 4 – Associate: care settings and prescribers

Priority cells are linked to the prescriber specialties and care settings that treat them. Drugs used in the indication are mapped to CMS Medicare Part D prescriber records; NPI taxonomy codes give specialty; volumes indicate where segments are concentrated. Open Payments data indicate where competitor engagement is already intense. For vaccines, care setting (paediatric, primary care, pharmacy) is taken from immunisation guidance. The output is a short list of specialties and settings per priority cell, with US-only coverage stated as a limitation.

### 5.10 Evaluation protocol

- **Back-test (E5):** features use evidence dated on or before 31 December 2019. A molecule–indication–segment is positive if it saw a Phase 2/3 start or an approval between 2020 and 2024 and had not reached that stage before the cut-off. Recall@k is primary, with Precision@k and NDCG@k [17] (k = 5, 10, 20). Leave-one-therapeutic-area-out validation tests generalisation.
- **Baselines:** random ranking; Open Targets score only; LLM-only segmentation and ranking without retrieval; heuristic multi-criteria score; learning-to-rank.
- **Ablation:** each viability dimension and each evidence source removed in turn.
- **Expert review (E3):** three to five domain experts rate segmentations and top-10 cells for five molecules, blinded to method (Likert 1–5).
- **Scale and pace (E6):** indications mapped per hour, LLM cost per molecule and analyst time to a validated draft versus manual assessment.
- **Statistics:** paired bootstrap 95% confidence intervals and Wilcoxon signed-rank tests.
- **Leakage controls:** label definition frozen before modelling; all evidence date-filtered; model versions and prompts pinned.

### 5.11 Prototype

A Streamlit application provides (a) a portfolio view of molecules with their top opportunity cells and scores, and (b) a drill-down showing each indication's opportunity assessment as a heat-map grid with segment sizes, priority tiers, associated specialties, cited evidence and SHAP drivers. Engineering practices include a single schema as the data contract, per-run persisted outputs and configuration snapshots, retry handling for LLM calls, unit tests with mocked LLM responses and versioned data snapshots.

### 5.12 Ethics, data governance and limitations

All data are open and public; no internal, confidential, licensed or personally identifiable patient data are used, and prescriber data are used only at aggregated specialty level. Source licences and terms of use (e.g., GBD, CMS, NLM) are respected and cited.

Limitations: development activity is a proxy for commercial opportunity; open data are richest for the US and sparser elsewhere; Medicare Part D covers mainly older patients; segment sizes rely on literature-derived ratios; and residual LLM hallucination is mitigated, not eliminated, by mandatory citation.

## 6. Plan of Work

The plan is aligned with the WILP dissertation calendar for the first semester of 2026–27. Figure 2 (Section 5.2) shows how each phase maps to the components of the system flow.

| Phase | Start Date – End Date | Work to be done |
|---|---|---|
| Literature review & outline | 24 Oct 2026 – 10 Nov 2026 | Literature review on market segmentation and AI methods; problem definition; submission of Dissertation Outline |
| Open-data acquisition & schema | 11 Nov 2026 – 30 Nov 2026 | Select molecule sample; ingest ChEMBL, Open Targets, AACT, PubMed, labels, GBD, CDC and CMS data; define schema and segmentation rules; annotation gold set. Abstract viva (20 Nov – 4 Dec) |
| Discover, extract & Segment | 01 Dec 2026 – 17 Dec 2026 | Indication discovery; LLM and transformer extraction (E1); knowledge graph; segmentation generator and validators (E3) |
| Mid-semester report | 18 Dec 2026 – 25 Dec 2026 | Prepare and submit mid-semester report |
| Evidence, Size & Align | 26 Dec 2026 – 20 Jan 2027 | Hybrid retrieval and Graph-RAG (E2); epidemiology sizing (E4); viability features, multi-criteria and learning-to-rank with SHAP. Mid-semester viva (7 – 21 Jan) |
| Associate, evaluation & prototype | 21 Jan 2027 – 30 Jan 2027 | CMS-based association; back-test and ablation (E5); expert review; scale study (E6); finalise prototype |
| Final dissertation | 31 Jan 2027 – 07 Feb 2027 | Supervisor and examiner review; write-up and submission |
| Final viva | 20 Feb 2027 – 05 Mar 2027 | Final presentation and viva |

## 7. Literature References

References are cited in the text by number in square brackets. The preliminary review covers market segmentation, retrieval-augmented generation and agentic LLMs, biomedical language models, knowledge graphs and drug repositioning, open pharmacology and epidemiology resources, and evaluation of grounded generation and ranking.


[1] M. Wedel and W. A. Kamakura, Market Segmentation: Conceptual and Methodological Foundations, 2nd ed. Boston, MA: Kluwer Academic Publishers, 2000.

[2] P. Lewis, E. Perez, A. Piktus, et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in Advances in Neural Information Processing Systems (NeurIPS), vol. 33, 2020.

[3] D. Edge, H. Trinh, N. Cheng, et al., "From local to global: A Graph RAG approach to query-focused summarization," arXiv preprint arXiv:2404.16130, 2024.

[4] S. Yao, J. Zhao, D. Yu, et al., "ReAct: Synergizing reasoning and acting in language models," in International Conference on Learning Representations (ICLR), 2023.

[5] J. Lee, W. Yoon, S. Kim, et al., "BioBERT: A pre-trained biomedical language representation model for biomedical text mining," Bioinformatics, vol. 36, no. 4, pp. 1234–1240, 2020.

[6] Y. Gu, R. Tinn, H. Cheng, et al., "Domain-specific language model pretraining for biomedical natural language processing," ACM Transactions on Computing for Healthcare, vol. 3, no. 1, 2021.

[7] K. Singhal, S. Azizi, T. Tu, et al., "Large language models encode clinical knowledge," Nature, vol. 620, pp. 172–180, 2023.

[8] D. S. Himmelstein, A. Lizee, C. Hessler, et al., "Systematic integration of biomedical knowledge prioritizes drugs for repurposing," eLife, vol. 6, e26726, 2017.

[9] S. Pushpakom, F. Iorio, P. A. Eyers, et al., "Drug repurposing: Progress, challenges and recommendations," Nature Reviews Drug Discovery, vol. 18, pp. 41–58, 2019.

[10] D. Ochoa, A. Hercules, M. Carmona, et al., "Open Targets Platform: Supporting systematic drug–target identification and prioritisation," Nucleic Acids Research, vol. 49, no. D1, pp. D1302–D1310, 2021.

[11] D. Mendez, A. Gaulton, A. P. Bento, et al., "ChEMBL: Towards direct deposition of bioassay data," Nucleic Acids Research, vol. 47, no. D1, pp. D930–D940, 2019.

[12] C. H. Wong, K. W. Siah, and A. W. Lo, "Estimation of clinical trial success rates and related parameters," Biostatistics, vol. 20, no. 2, pp. 273–286, 2019.

[13] A. Tasneem, L. Aberle, H. Ananth, et al., "The database for Aggregate Analysis of ClinicalTrials.gov (AACT) and subsequent regrouping by clinical specialty," PLoS ONE, vol. 7, no. 3, e33677, 2012.

[14] GBD 2019 Diseases and Injuries Collaborators, “Global burden of 369 diseases and injuries in 204 countries and territories, 1990–2019,” The Lancet, vol. 396, pp. 1204–1222, 2020.

[15] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, “RAGAS: Automated evaluation of retrieval augmented generation,” in Proceedings of EACL 2024: System Demonstrations, 2024.

[16] C. J. C. Burges, “From RankNet to LambdaRank to LambdaMART: An overview,” Microsoft Research Technical Report MSR-TR-2010-82, 2010.

[17] K. Järvelin and J. Kekäläinen, “Cumulated gain-based evaluation of IR techniques,” ACM Transactions on Information Systems, vol. 20, no. 4, pp. 422–446, 2002.

[18] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in Advances in Neural Information Processing Systems (NeurIPS), vol. 30, 2017.

## 8. Particulars of the Supervisor and Additional Examiner

| | **Supervisor** | **Additional Examiner** |
|---|---|---|
| Name | Anurag Sinha | Sangitha V |
| Qualification | MBA, Business Administration, Management and Operations – Indian Institute of Management<br>Bachelor's, Business Administration and Management – Indian Institute of Management | B.Tech. Biotechnology |
| Designation | Senior Manager, Brand Strategy and Analytics | Senior Analyst, Forecasting |
| Employing Organization and Location | MSD Hyderabad Global Capabilities Center India Private Limited, Hyderabad | Sandoz Private Ltd |
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## 9. Remarks of the Supervisor

The proposed dissertation addresses a real and growing need in brand strategy and analytics assessing the commercial viability of many molecules quickly and consistently by producing sized, prioritized and evidence-backed draft opportunity assessments before detailed manual validation begins. The problem is well defined, the scope is appropriate for a Master's dissertation, and the exclusive use of open public data keeps the work independent of confidential information. The student has reviewed relevant literature on market segmentation, retrieval-augmented generation, biomedical language models, knowledge graphs and ranking methods, and has set clear evaluation metrics, including retrospective back-testing, sizing accuracy, expert review and a scale study. Given his experience in commercial analytics and forecasting, I am confident that the student can complete the work in the planned timeline. I approve the proposed dissertation as the supervisor.

**Signature of Supervisor:** Anurag Sinha

**Name: Anurag Sinha**

**Date: _____**

**BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI**

**WORK INTEGRATED LEARNING PROGRAMMES (WILP) DIVISION**

**FIRST SEMESTER OF ACADEMIC YEAR 2026-2027**

**AIMLCZG628T : DISSERTATION OUTLINE**

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